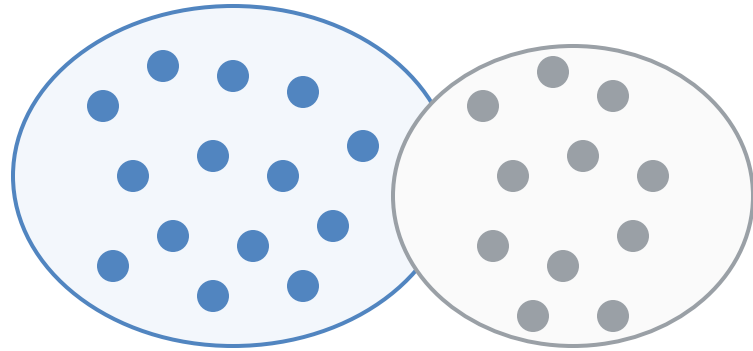


1 Small pilot assay

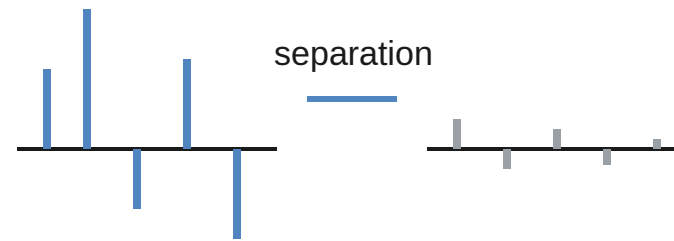
pilot perturbation cells WT / control



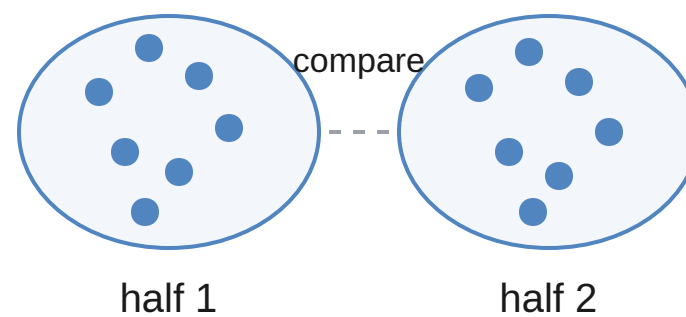
25-50 cells per perturbation

2 Estimate measurement properties

variant-to-WT effect size



within-variant split-half noise



pilot effect-size-to-noise
estimate

3 Predict rankability

likely to clear the
measurement floor?

likely rankable

proceed to full-depth
benchmarking

likely below floor

increase depth,
redesign, or report as
measurement-limited

predicts benchmark eligibility, not expression response
or downstream model performance

Can a small pilot identify perturbations whose ground truth is likely to become measurable at higher depth?